

PROFILE

Applied AI and platform engineer with about eight years of experience turning enterprise requirements into production systems, including agent application delivery, model integration, access control, rollout, and reliability engineering.

Builds hands-on across Python application logic, cloud-native runtime integration, security controls, rollout, and production operations.

SELECTED EVIDENCE

Governed AX application delivery with existing enterprise LLM Gateway integration

Public assistant reference with retrieval, RBAC, citations, audit, and evals

UK rollout and production stabilization for an Azure-based generative AI service

CORE SKILLS

Applied AI	Agent application delivery, retrieval, grounded citations, evaluation cases, model inference, LLM Gateway API integration
Software	Python, FastAPI, SQLite FTS5, Bash, Linux, API and runtime integration, asynchronous flows
Cloud Platform	AWS EKS, Azure, Kubernetes, OpenShift, Docker, Helm/Helmfile, ArgoCD
Production Quality	Authentication, authorization, governance, observability, incident response, rollout stabilization

EXPERIENCE

Samsung Electronics, AX Development Group

Seoul, South Korea

SRE / AI Platform Engineer

Jan. 2026 - Present

- Design and build an internal AX Agent Platform on AWS EKS, adapting the agent runtime and deployment path to enterprise platform standards.
- Implemented self-service AX application delivery with automatic domain issuance, authorized-only access, and platform controls for authentication, authorization, security, and governance.
- Enabled AX applications to call the existing enterprise LLM Gateway API through platform-side client compatibility and runtime integration.
- Supported a Microsoft-collaborative B2C generative AI service through UK rollout readiness, initial production operations, incident response, and stabilization from Jan. to Jun. 2026.

Samsung Research

Seoul, South Korea

SRE / AI Platform Engineer

Jul. 2024 - Dec. 2025

- Led production delivery and application reliability for a company-wide AI chatbot serving 30,000 employees on internal Kubernetes clusters.
- Owned releases across development, staging, and production while integrating application delivery with security and operational requirements.
- Built container delivery pipelines with Docker, Kubernetes, GitHub Actions, ArgoCD, Helm, and Helmfile.
- Deployed Grafana, Prometheus, and OpenTelemetry observability for shared service visibility and operational diagnosis.

Samsung Electronics, Network Division

Seoul, South Korea

SRE & Software Engineer

Jan. 2021 - Jun. 2024

- Designed and delivered containerized applications with Docker, Helm, Kubernetes Operators, and AWS services including EKS, ECS, Route 53, RDS, and IAM.
- Built and managed OpenShift, Kubernetes, and EKS environments for reliability, high availability, performance, and auto-scaling.
- Designed and developed a Bash-based one-click Kubernetes cluster provisioning system for cloud developers.
- Developed a Linux application that mirrors and forwards up to 130,000 packets per second under constrained system resources.

EXPERIENCE (CONTINUED)

TmaxOS & TmaxCloud

Gyeonggi, South Korea

Software & Platform Engineer

Jan. 2019 - Dec. 2020

- Built a Siri-like AI assistant on TmaxOS, covering model inference, data preprocessing, system integration, and asynchronous application flows.
- Designed and developed Kubernetes CI/CD and registry operators and deployed a private container image registry.

SELECTED ENGINEERING WORK

Enterprise AI Assistant | Public reference implementation

jinwonlee.github.io/projects/enterprise-ai-assistant/web/

Built an access-aware Python assistant with SQLite FTS5 retrieval, pre-generation role filtering, grounded citations, audit events, evaluation cases, a FastAPI surface, and an optional OpenAI Responses API provider.

TmaxOS AI Assistant

Built model inference, preprocessing, system integration, and asynchronous interaction flows for an OS-level AI assistant.

Distributed CNN Training Research

Studied resource-aware placement for distributed TensorFlow training and contributed to research artifacts reporting up to 2.3x training speedup.

EDUCATION & RESEARCH

UNIST | M.S. in Computer Science and Engineering

Mar. 2017 - Feb. 2019

Master's thesis: Convergence Aware CNN Training | Advisors: Jiwon Seo and Sam H. Noh

UNIST | B.S. in Computer Science and Engineering, Minor in Mathematics

Mar. 2012 - Feb. 2017

Undergraduate research: distributed systems for AI model training with TensorFlow

PUBLICATIONS & LANGUAGE

- USENIX NSDI 2019 poster: Alleviating the Network Bottleneck for CNN Distributed Training through Automatic Resource-Aware Layer Placement.
- arXiv 2019: Accelerated Training for CNN Distributed Deep Learning through Automatic Resource-Aware Layer Placement.
- ACM EuroSys 2018 poster: Improving Performance of Distributed TensorFlow using CNN Characteristics Exploiting Model Parallelism.
- OPIc IH (Intermediate High), Dec. 2025.